

STAREHE GIRLS' 2025 PREDICTION



POSSIBLE KCSE EXAM



451/1

COMPUTER STUDIES

PAPER 1 (THEORY)

TIME: 2½ HOURS

NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

(Kenya Certificate of Secondary Education)

INSTRUCTIONS TO CANDIDATES

- (a)** Write your Name and Index/Adm Number in the spaces provided.
- (b)** This paper consists of **TWO** sections: **A** and **B**
- (c)** Answer **ALL** the questions in section **A**
- (d)** Answer question **16** and any other **THREE** questions from section **B**
- (e)** All answers should be written in the spaces provided in the question paper.
- (f)** Candidates should answer the questions in English.

SECTION A: 40MKS

Answer **ALL** the questions in this section

1. Name four categories of input devices

(4marks)

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2. State any two data transmission impairments

(2marks)

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3. Explain the emerging trends in microcomputer technology in relation to size **(2marks)**

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4. Name any two physical measures taken to protect a computer lab from unauthorized access.

(2 marks)

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5. Mention two things that are likely to cause strain- in the computer room (2 marks)

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6. What are turnaround documents? (1 mark)

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7. Name any two data capture techniques that make use of turnaround documents (2 marks)

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8. State four functions of Read – Only Memory (ROM) (4 marks)

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9. Lydiah connected new multimedia speakers to her computer and tried to play her favorite music CD, but no sound came out. Suggest two problems that might have occurred (2 marks)

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10. Explain why documentation is necessary for any software package (2 marks)

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11. A firm intends to purchase new software. List three items of documentation that should accompany the software (2 marks)

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12. Give three advantages of using GUI based operating system over a command line interface

(3marks)

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13. Distinguish between disk compressing and disk defragmenting

(2 marks)

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14. Describe TWO roles of each of the following;

(a) Database Administrator

(2 marks)

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b) System analyst

(2marks)

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15. Define the following terms in relation to computer security: (6 marks)

a) Data Encryption

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b) Sabotage

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c) Firewall

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SECTION B: 60mks

Answer question 16 (Compulsory) and any other THREE questions from this section

16. a) State the three basic types of program control structures. (3 marks)

b) Below is an algorithm that is used to compute the values of R, S and T.

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P = 5
Q = 6
INPUT N
If N is GREATER OR EQUAL TO 10
R = P * Q
S = Q - P
T = P + Q + R + S
ELSE
R = P + Q
S = Q
T = R + S
END IF
PRINT R, S and T

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c) From the algorithm, determine the output if the input value of N is (6 marks)

i) 7;

ii) 10;

d) Draw a flowchart for the algorithm in question 16(b) (6 marks)

17. a) Name and describe Three main application areas of Artificial Intelligence in ICT.

(6 marks)

b) State three advantages of automated production in manufacturing industries (3 marks)

c) List FOUR types of computer processing files. (4 marks)

d) Distinguish between data verification and data validation (2 marks)

18. a) Name and Explain any three types of errors that are likely to exist in a program?

(6 marks)

b) Explain TWO functions of networking software in a computer (2 marks)

c) Give two advantages and two disadvantages of impact printers (4 marks)

d) State three features of a fifth generation computer that distinguishes it from a fourth generation computer. (3 mark)

19. a) Explain the following types of data processing modes

(i) Batch processing **(2 marks)**

(ii) Real time processing **(2 marks)**

b) Differentiate between phased and parallel implementation methods. **(2marks)**

c) A student presented a budget in the form of a worksheet as follows

	A	B	C
1	Item	Amount	
2	Fare	200	
3	Stationery	50	
4	Bread	300	
5	Miscellaneous	150	
6	Total		

i) Given that the value 0.5 is typed in cell B9, write the shortest formula that would be typed in cell C2 and then copied down the column to obtain half the values in column B.

(2 marks)

ii) Write two different formulae that can be typed to obtain the total in cell B6 and then copied to cell C6.

(2 marks)

d) (i) Give any two components of Satellite. **(2 marks)**

(ii) Give three advantages of satellite. **(3 marks)**

20. a) Describe the mesh topology as used in networking. **(2 marks)**

b) Give three Advantages of the mesh topology. **(3 marks)**

c) Give two Examples of radio waves. **(2 marks)**

d) Give two reasons why would you prefer fiber optical cables to any other cable? (2 marks)

e) State two reasons for use of binary in digital technology. (2 marks)

f) Determine the value of X in the equations:

i) $X_2 + 1101_2 = 70_{10}$ (2 marks)

ii) $100110_2 - X_2 = 001010_2$ (2 marks)